Yash Agarwal

Embedded Firmware Engineer

RESUME_1014

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Total Experience: 12 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 12 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: UART, C, Git, SPI, Firmware Debugging, ARM Cortex-M, FreeRTOS, C++, Oscilloscope, CAN, STM32, JTAG

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | OrionPath Technologies | Jan 2022 - Present

- Reduced embedded task jitter by 39% using interrupt priority and queue tuning
- Validated 578 hardware test vectors across protocol and timing corner cases
- Improved battery runtime by 14% with peripheral and sleep-state optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | ZenithTrail Technologies | Jan 2018 - Dec 2021

- Improved battery runtime by 36% with peripheral and sleep-state optimization
- Lowered field-reset incidents by 13% with watchdog recovery and diagnostics

Associate Embedded Firmware Engineer | KoraLink Innovations | Jul 2014 - Dec 2017

- Reduced embedded task jitter by 41% using interrupt priority and queue tuning
- Validated 154 hardware test vectors across protocol and timing corner cases

PROJECTS

Industrial Sensor Firmware | Oscilloscope, UART, CAN, Git

- Built FreeRTOS tasks for calibrated sampling and watchdog recovery.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | STM32, UART, Oscilloscope, SPI

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Crescent Valley Technical University, 2014
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

Embedded C Standards (2026)